person's heart by attaching electrodes to their chest and arms (an electrocardiogram or EKG). The *shape* of the EKG waveform provides the information being sought, such as when the various chambers contract during a heartbeat. Images are another example of this type of signal. Rather than a waveform that varies over *time*, images encode information in the shape of a waveform that varies over *distance*. Pictures are formed from regions of brightness and color, and how they relate to other regions of brightness and color. You don't look at the *Mona Lisa* and say, *"My, what an interesting collection of sinusoids."*

Here's the problem: The sampling theorem is an analysis of what happens in the frequency domain during digitization. This makes it ideal to under-stand the analog-to-digital conversion of signals having their information encoded in the frequency domain. However, the sampling theorem is little help in understanding how time domain encoded signals should be digitized. Let's take a closer look.

Figure 3-15 illustrates the choices for digitizing a time domain encoded signal. Figure (a) is an example analog signal to be digitized. In this case, the information we want to capture is the *shape* of the rectangular pulses. A short burst of a high frequency sine wave is also included in this example signal. This represents wideband noise, interference, and similar junk that always appears on analog signals. The other figures show how the digitized signal would appear with different antialias filter options: a Chebyshev filter, a Bessel filter, and no filter.

It is important to understand that *none* of these options will allow the original signal to be reconstructed from the sampled data. This is because the original signal inherently contains frequency components greater than one-half of the sampling rate. Since these frequencies cannot exist in the digitized signal, the reconstructed signal cannot contain them either. These high frequencies result from two sources: (1) noise and interference, which you would like to eliminate, and (2) sharp edges in the waveform, which probably contain information you want to retain.

The Chebyshev filter, shown in (b), attacks the problem by aggressively removing all high frequency components. This results in a filtered analog signal that *can* be sampled and later perfectly reconstructed. However, the reconstructed analog signal is identical to the *filtered signal*, not the *original signal*. Although nothing is lost in sampling, the waveform has been severely distorted by the antialias filter. As shown in (b), the cure is worse than the disease! Don't do it!

The Bessel filter, (c), is designed for just this problem. Its output closely resembles the original waveform, with only a gentle rounding of the edges. By adjusting the filter's cutoff frequency, the smoothness of the edges can be traded for elimination of high frequency components in the signal. Using more poles in the filter allows a *better* tradeoff between these two parameters. A common guideline is to set the cutoff frequency at about one-quarter of the sampling frequency. This results in about two samples